Partial Differential Equations

Yuchao Feng

fengyuchao@sjtu.edu.cn

Contents

1 Introduction 1
1.1 Basic Concepts 1
1.2 Typical Equations 2
1.3 Second-Order Linear Equations 3
1.4 Characteristics 7
2 Wave Equations 9
2.1 First-Order Wave Equations 9
2.2 Higher-Order Wave Equations 12

1 Introduction

1.1 Basic Concepts

Definition 1.1. *(Partial Differential Equation)*
Let $x = (x_1, \cdots, x_n)$ and let $u = u(x)$ be the unknown function. Write

$$Du = \left(u_{x_1}, \cdots, u_{x_n}\right), \quad D^k u = \left(\frac{\partial^k u}{\partial x_1^{k_1} \cdots \partial x_n^{k_n}}\right)_{k_1 + \cdots + k_n = k}.$$

An equation

$$F(x, u, Du, D^2 u, \cdots, D^N u) = f(x)$$

is a partial differential equation of order N, denoted by $Gu = f(x)$.

Definition 1.2. *(Multi-index)*
A multi-index is $\alpha = (\alpha_1, \cdots, \alpha_n)$ with length $|\alpha| = \alpha_1 + \cdots + \alpha_n$ and

$$\partial^\alpha = \frac{\partial^{|\alpha|}}{\partial x_1^{\alpha_1} \cdots \partial x_n^{\alpha_n}}.$$

Definition 1.3. *(Homogeneous Equation)*
The equation $Gu = f(x)$ is homogeneous when $f \equiv 0$, and nonhomogeneous when $f \neq 0$.

Definition 1.4. *(Linear Equation)*
The equation is linear when $G(au + bv) = aGu + bGv$ for constants a, b. In multi-index form,

$$\sum_{|\alpha| \leq N} A_\alpha(x)\partial^\alpha u = f(x).$$

Otherwise the equation is nonlinear.

Definition 1.5. *(Semilinear, Quasilinear, and Fully Nonlinear Equations)*
The principal part is the collection of terms that contain a derivative of order N.

- The equation is semilinear when the principal part is linear, with coefficients depending only on x:

$$\sum_{|\alpha| = N} A_\alpha(x)\partial^\alpha u + A_0(x, u, Du, \cdots, D^{N-1}u) = f(x).$$

- The equation is quasilinear when it is linear in the derivatives of order N, whose coefficients may depend on derivatives of lower order:

$$\sum_{|\alpha| = N} A_\alpha(x, u, Du, \cdots, D^{N-1}u)\partial^\alpha u + A_0(x, u, Du, \cdots, D^{N-1}u) = f(x).$$

- The equation is fully nonlinear when a derivative of order N enters nonlinearly.

1.2 Typical Equations

Definition 1.6. *(The Wave Equation)*
Let $a > 0$ and let $\Delta = \partial_{x_1 x_1} + \cdots + \partial_{x_n x_n}$. The wave equation is

$$u_{tt} - a^2 \Delta u = f$$

with initial conditions

$$u(x, 0) = \varphi(x), \quad u_t(x, 0) = \psi(x).$$

Definition 1.7. *(The Heat Equation)*
Let $a > 0$. The heat equation, also called the diffusion equation, is

$$u_t - a^2 \Delta u = f$$

with initial conditions

$$u(x,0) = \varphi(x).$$

Definition 1.8. *(Laplace's Equation)*
Laplace's equation is $\Delta u = 0$, and a solution of Laplace's equation is harmonic. The equation does not involve time, so there is no initial condition.

Definition 1.9. *(Boundary Conditions)*
Let $\Gamma = \partial\Omega$, let $\boldsymbol{n}$ be the outward unit normal, and let $\sigma > 0$. Dirichlet, Neumann, and Robin conditions are respectively given by

$$u|_\Gamma = g, \quad \left.\frac{\partial u}{\partial \boldsymbol{n}}\right|_\Gamma = g, \quad \left.\left(\frac{\partial u}{\partial \boldsymbol{n}} + \sigma u\right)\right|_\Gamma = g.$$

1.3 Second-Order Linear Equations

Definition 1.10. *(Discriminant and Type)*
Let $a_{11}^2 + a_{12}^2 + a_{22}^2 \neq 0$, and let $u = u(x,y)$ satisfy

$$a_{11}u_{xx} + 2a_{12}u_{xy} + a_{22}u_{yy} + F(x,y,u,Du) = 0.$$

The discriminant is $\Delta = a_{12}^2 - a_{11}a_{22}$, where
- $\Delta > 0$: the equation is hyperbolic;
- $\Delta = 0$: the equation is parabolic;
- $\Delta < 0$: the equation is elliptic.

Remark.
From here on the equation is linear,

$$Lu = a_{11}u_{xx} + 2a_{12}u_{xy} + a_{22}u_{yy} + b_1 u_x + b_2 u_y + cu = f,$$

with coefficients depending on (x,y). When a change of independent variables $\xi = \xi(x,y), \eta = \eta(x,y)$ is invertible near (x_0, y_0), the chain rule gives

$$u_x = u_\xi \xi_x + u_\eta \eta_x, \quad u_y = u_\xi \xi_y + u_\eta \eta_y,$$

$$u_{xx} = u_{\xi\xi}\xi_x^2 + 2u_{\xi\eta}\xi_x\eta_x + u_{\eta\eta}\eta_x^2 + u_\xi \xi_{xx} + u_\eta \eta_{xx},$$

$$u_{xy} = u_{\xi\xi}\xi_x\xi_y + u_{\xi\eta}\big(\xi_x\eta_y + \xi_y\eta_x\big) + u_{\eta\eta}\eta_x\eta_y + u_\xi\xi_{xy} + u_\eta\eta_{xy},$$

$$u_{yy} = u_{\xi\xi}\xi_y^2 + 2u_{\xi\eta}\xi_y\eta_y + u_{\eta\eta}\eta_y^2 + u_\xi\xi_{yy} + u_\eta\eta_{yy}.$$

Substitution produces

$$\overline{a}_{11}u_{\xi\xi} + 2\overline{a}_{12}u_{\xi\eta} + \overline{a}_{22}u_{\eta\eta} + \overline{b}_1u_\xi + \overline{b}_2u_\eta + \overline{c}u = \overline{f},$$

with $\overline{c} = c$, $\overline{f} = f$, and

$$\overline{a}_{11} = a_{11}\xi_x^2 + 2a_{12}\xi_x\xi_y + a_{22}\xi_y^2,$$

$$\overline{a}_{12} = a_{11}\xi_x\eta_x + a_{12}\big(\xi_x\eta_y + \xi_y\eta_x\big) + a_{22}\xi_y\eta_y,$$

$$\overline{a}_{22} = a_{11}\eta_x^2 + 2a_{12}\eta_x\eta_y + a_{22}\eta_y^2,$$

$$\overline{b}_1 = a_{11}\xi_{xx} + 2a_{12}\xi_{xy} + a_{22}\xi_{yy} + b_1\xi_x + b_2\xi_y,$$

$$\overline{b}_2 = a_{11}\eta_{xx} + 2a_{12}\eta_{xy} + a_{22}\eta_{yy} + b_1\eta_x + b_2\eta_y.$$

There is $\overline{\Delta} = |J|^2\ \Delta$ with $J = \begin{pmatrix}\xi_x & \xi_y\\ \eta_x & \eta_y\end{pmatrix}$, so the type is invariant.

Lemma 1.1.
If φ solves the principal equation above and $\varphi_x^2 + \varphi_y^2 \neq 0$, then

$$a_{11}(\mathrm{d}y)^2 - 2a_{12}\,\mathrm{d}x\,\mathrm{d}y + a_{22}(\mathrm{d}x)^2 = 0.$$

Proposition 1.1. *(Canonical Forms for Constant Coefficients)*
Suppose a_{ij}, b_j, and c are constant. A linear change of (x, y) puts the equation in one of the following forms, with constant coefficients on the right-hand side.

- Hyperbolic (first canonical form): $u_{\xi\eta} = Au_\xi + Bu_\eta + Cu + D$.
- Hyperbolic (second canonical form): $u_{rr} - u_{ss} = Au_r + Bu_s + Cu + D$.
- Parabolic: $u_{\eta\eta} = Au_\xi + Bu_\eta + Cu + D$.
- Elliptic: $u_{rr} + u_{ss} = Au_r + Bu_s + Cu + D$.

Example.
The equation $4u_{xx} + 5u_{xy} + u_{yy} + u_x + u_y = 2$ has $\Delta = \left(\frac{5}{2}\right)^2 - 4 > 0$, so it is hyperbolic. The characteristic equation $4(\mathrm{d}y)^2 - 5\,\mathrm{d}x\,\mathrm{d}y + (\mathrm{d}x)^2 = 0$ factors as $(4\,\mathrm{d}y - \mathrm{d}x)(\mathrm{d}y - \mathrm{d}x) = 0$, with characteristic lines $4y - x = C_1$ and $y - x = C_2$.

Set $\xi = 4y - x$ and $\eta = y - x$. Then $\overline{a}_{11} = \overline{a}_{22} = 0$, $\overline{a}_{12} = -\frac{9}{2}$, $\overline{b}_1 = 3$, and $\overline{b}_2 = 0$, so $-9u_{\xi\eta} + 3u_\xi = 2$, that is

$$u_{\xi\eta} = \frac{1}{3}u_\xi - \frac{2}{9}.$$

Writing $w = u_\xi$ gives $w_\eta - \frac{w}{3} = -\frac{2}{9}$. Multiplication by $e^{-\frac{\eta}{3}}$ produces $w = \frac{2}{3} + F(\xi)e^{\frac{\eta}{3}}$, and one further integration gives

$$u(x,y) = \frac{2}{3}(4y - x) + f(4y - x)e^{\frac{y-x}{3}} + g(y - x).$$

Example.
The equation $u_{xx} - 4u_{xy} + 4u_{yy} = e^y$ has $\Delta = 2^2 - 4 = 0$, so it is parabolic. The characteristic equation $(\mathrm{d}y + 2\,\mathrm{d}x)^2 = 0$ has the lines $y + 2x = C$. Set $\xi = y + 2x$ and $\eta = y$. Then $\overline{a}_{11} = \overline{a}_{12} = 0$, $\overline{a}_{22} = 4$, and the lower-order coefficients coming from ξ and η vanish, so

$$u_{\eta\eta} = \frac{1}{4}e^\eta.$$

Two integrations give $u = \frac{1}{4}e^\eta + \eta f(\xi) + g(\xi)$, hence

$$u(x,y) = \frac{1}{4}e^y + yf(y + 2x) + g(y + 2x).$$

Example.
The equation $u_{xx} + u_{xy} + u_{yy} + u_x = 0$ has $a_{12} = \frac{1}{2}$ and $\Delta = \left(\frac{1}{2}\right)^2 - 1 < 0$, so it is elliptic. The characteristic equation $\mathrm{d}y^2 - \mathrm{d}x\,\mathrm{d}y + \mathrm{d}x^2 = 0$ has roots

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{2} \pm \frac{\sqrt{3}}{2}i.$$

Set $\xi = y - \frac{x}{2}$ and $\eta = -\frac{\sqrt{3}}{2}x$. Then

$$\overline{a}_{11} = \overline{a}_{22} = \frac{3}{4}, \quad \overline{a}_{12} = 0, \quad \overline{b}_1 = -\frac{1}{2}, \quad \overline{b}_2 = -\frac{\sqrt{3}}{2},$$

and the equation becomes

$$u_{\xi\xi} + u_{\eta\eta} = \frac{2}{3}u_\xi + \frac{2}{\sqrt{3}}u_\eta.$$

Example.
The equation $y^2u_{xx} - x^2u_{yy} = 0$ has $\Delta = x^2y^2$, so it is hyperbolic off the coordinate axes. The characteristic equation $y^2(\mathrm{d}y)^2 - x^2(\mathrm{d}x)^2 = 0$ factors as $(y\,\mathrm{d}y + x\,\mathrm{d}x)(y\,\mathrm{d}y - x\,\mathrm{d}x) = 0$, with integrals

$$\frac{1}{2}y^2 - \frac{1}{2}x^2 = C_1, \quad \frac{1}{2}y^2 + \frac{1}{2}x^2 = C_2.$$

Set $\xi = \frac{y^2-x^2}{2}$ and $\eta = \frac{y^2+x^2}{2}$. Then, wherever $|\xi| \neq |\eta|$,

$$u_{\xi\eta} = \frac{\eta}{2(\xi^2 - \eta^2)}u_\xi - \frac{\xi}{2(\xi^2 - \eta^2)}u_\eta.$$

Example.
The equation $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} + xyu_x + y^2u_y = 0$ has $\Delta = 0$, so it is parabolic. The characteristic equation $(x\,\mathrm{d}y - y\,\mathrm{d}x)^2 = 0$ has the lines $\frac{y}{x} = C$. Set $\xi = \frac{y}{x}$ and $\eta = y$. Then $u_{\eta\eta} = -u_\eta$. Writing $w = u_\eta$ and integrating twice gives

$$u(x,y) = -e^{-y}f\left(\frac{y}{x}\right) + g\left(\frac{y}{x}\right).$$

Example.
The equation $u_{xx} + x^2u_{yy} = 0$ has $\Delta = -x^2$, so it is elliptic off the axis $x = 0$. The characteristic equation $\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2 = -x^2$ has complex integrals $2y \pm ix^2 = C$. Set $\xi = 2y$ and $\eta = x^2$. Then, for $\eta \neq 0$,

$$u_{\xi\xi} + u_{\eta\eta} = -\frac{1}{2\eta}u_\eta.$$

Proposition 1.2. *(Reduced Canonical Forms)*
For constant coefficients, first-order terms may be removed by the exponential substitution $u = ve^{ar+bs}$. Starting from the first canonical forms and choosing the right a and b, this produces the reduced forms

$$v_{rs} = Ev + F, \quad v_{rr} \pm v_{ss} = Ev + F.$$

Definition 1.11. *(Second-Order Linear Operator)*
For $x = (x_1, \cdots, x_n)$ and a symmetric matrix $A = \big(a_{ij}\big)$,

$$Lu = \sum_{i,j=1}^{n} a_{ij} u_{x_i x_j} + \sum_{j=1}^{n} b_j u_{x_j} + cu = f.$$

The type at a point is read from the eigenvalues $\lambda_1, \cdots, \lambda_n$ of A.

Definition 1.12. *(Types in Several Variables)*
Let $\lambda_1, \cdots, \lambda_n$ be the eigenvalues of A.

- Hyperbolic: every λ_i is nonzero, $n-1$ of them have the same sign, and one has the opposite sign.
- Parabolic: exactly one eigenvalue is zero and the others have the same sign.
- Elliptic: every λ_i is nonzero and all have the same sign.
- Ultrahyperbolic: for $n \geq 4$, every λ_i is nonzero, at least two are positive, and at least two are negative.
- Ultraparabolic: more than one eigenvalue is zero and the remaining eigenvalues have the same sign.

1.4 Characteristics

Definition 1.13. *(Characteristic Direction)*
At a fixed point x, a direction $\boldsymbol{l} = (\alpha_1, \cdots, \alpha_n)$ is a characteristic direction if it satisfies the characteristic equation

$$\sum_{i,j=1}^{n} a_{ij}(x) \alpha_i \alpha_j = 0.$$

Definition 1.14. *(Characteristic Surface)*
The hypersurface $S : \varphi(x) = 0$ is a characteristic surface if at every point of S the normal $\left(\varphi_{x_1}, \cdots, \varphi_{x_n}\right)$ is a characteristic direction, namely

$$\sum_{i,j=1}^{n} a_{ij}(x) \varphi_{x_i} \varphi_{x_j} = 0.$$

Remark.
To find characteristic surfaces, solve the first-order equation above for φ. When $n = 2$, along $\varphi = C$ one has $\varphi_x \, \mathrm{d}x + \varphi_y \, \mathrm{d}y = 0$, hence $\mathrm{d}y / \mathrm{d}x = -\varphi_x / \varphi_y$. Substituting the normal $\left(\varphi_x, \varphi_y\right)$ into the characteristic equation gives the ODE

$$a_{11} (\mathrm{d}y)^2 - 2a_{12} \, \mathrm{d}x \, \mathrm{d}y + a_{22} (\mathrm{d}x)^2 = 0$$

used to reduce the two-variable equation to canonical form. In two variables a characteristic surface is a characteristic curve.

Definition 1.15. *(Characteristic Plane)*
A characteristic plane is an $(n-1)$-dimensional hyperplane whose normal is a characteristic direction. Through a point $\overline{x}$ with characteristic direction $(\alpha_1, \cdots, \alpha_n)$ it is

$$\sum_{i=1}^{n} \alpha_i (x_i - \overline{x}_i) = 0.$$

Definition 1.16. *(Characteristic Cone)*
The characteristic cone at a fixed point $\overline{x}$ is the cone formed by the envelope of all characteristic planes through $\overline{x}$.

Remark.
If the characteristic planes through $\overline{x}$ form a family $F(x, \lambda) = 0$, the envelope is obtained by eliminating the parameters λ from $F = 0$ and the equations obtained by differentiating F with respect to each parameter. Away from the vertex, the cone is itself a characteristic surface.

Example.
For Laplace's equation $\Delta u = 0$, the characteristic equation is $\alpha_1^2 + \cdots + \alpha_n^2 = 0$. There is no real characteristic direction, hence no real characteristic surface.

Example.
For the heat equation $u_t = a^2\left(u_{xx} + u_{yy} + u_{zz}\right)$, write $\boldsymbol{l} = (\alpha_0, \alpha_1, \alpha_2, \alpha_3)$ with $|\boldsymbol{l}| = 1$. The principal part involves only the spatial second derivatives, so the characteristic equation is $\alpha_1^2 + \alpha_2^2 + \alpha_3^2 = 0$. Together with $|\boldsymbol{l}| = 1$ this forces $\alpha_0 = \pm 1$ and $\alpha_1 = \alpha_2 = \alpha_3 = 0$. The characteristic directions are $\boldsymbol{l} = (\pm 1, 0, 0, 0)$, and the characteristic planes are $t = \text{const}$. Through a given point there is only one such plane, so the characteristic cone degenerates to that plane.

Example.
For the wave equation $u_{tt} = a^2\left(u_{xx} + u_{yy}\right)$, the characteristic equation $\alpha_0^2 = a^2(\alpha_1^2 + \alpha_2^2)$ together with $|\boldsymbol{l}| = 1$ gives

$$\alpha_0 = \pm\frac{a}{\sqrt{1+a^2}}, \quad \alpha_1^2 + \alpha_2^2 = \frac{1}{1+a^2}.$$

Parameterizing the spatial part yields the characteristic directions

$$\boldsymbol{l} = \left(\pm\frac{a}{\sqrt{1+a^2}}, \frac{\cos\varphi}{\sqrt{1+a^2}}, \frac{\sin\varphi}{\sqrt{1+a^2}}\right).$$

If θ is the angle between $\boldsymbol{l}$ and the t-axis, then $\tan\theta = 1/a$. Scaling the normal, the characteristic planes through (t_0, x_0, y_0) are

$$a(t-t_0) + \cos\varphi\cdot(x-x_0) + \sin\varphi\cdot(y-y_0) = 0.$$

Differentiating with respect to φ yields $-\sin\varphi\cdot(x-x_0) + \cos\varphi\cdot(y-y_0) = 0$. Squaring and adding these two equations eliminates φ and produces the characteristic cone

$$(x-x_0)^2 + (y-y_0)^2 = a^2(t-t_0)^2.$$

2 Wave Equations

2.1 First-Order Wave Equations

Theorem 2.1. *(d'Alembert's Formula)*
If $\varphi \in C^2(\mathbb{R})$ and $\psi \in C^1(\mathbb{R})$, the Cauchy problem

$$u_{tt} - a^2 u_{xx} = 0, \quad u(x,0) = \varphi(x), \quad u_t(x,0) = \psi(x)$$

has a unique classical solution

$$u(x,t) = \frac{\varphi(x-at) + \varphi(x+at)}{2} + \frac{1}{2a}\int_{x-at}^{x+at} \psi(y)\,\mathrm{d}y.$$

The solution depends continuously on the initial data: if $\|\varphi_1 - \varphi_2\|_\infty < \varepsilon$ and $\|\psi_1 - \psi_2\|_\infty < \varepsilon$, then

$$\|u_1(\cdot,t) - u_2(\cdot,t)\|_\infty \le \varepsilon(1+t).$$

Proof.
The change $\xi = x - at$, $\eta = x + at$ reduces the equation to $u_{\xi\eta} = 0$, hence $u(x,t) = F(x-at) + G(x+at)$. The initial conditions give $F(x) + G(x) = \varphi(x)$ and $-aF'(x) + aG'(x) = \psi(x)$. Integrating the second identity and solving for F and G produces the formula. Any classical solution is of this form, so the solution is unique.

The stability bound follows by estimating each term, using that the ψ-integral has length $2at$. ■

Example.
For $u_{tt} - u_{xx} = 0$ with $\varphi(x) = \frac{1}{1+x^2}$ and $\psi = 0$,

$$u(x,t) = \frac{1}{2}\left(\frac{1}{1+(x+t)^2} + \frac{1}{1+(x-t)^2}\right), \quad \lim_{t\to+\infty} u(x,t) = 0.$$

Example.
For $u_{tt} - u_{xx} = 0$ with $\varphi = 0$ and $\psi(x) = \frac{2}{1+x^2}$,

$$u(x,t) = \arctan(x+t) - \arctan(x-t), \quad \lim_{t\to+\infty} u(x,t) = \pi.$$

Proposition 2.1. *(Invariance)*
If u solves the homogeneous equation, then so do $u(x-y,t)$ for fixed y, the space and time reversals of u, the rescaling $u(kx,kt)$, every derivative of u, and the convolution $\int_a^b S(x-y,t)g(y)\,\mathrm{d}y$ whenever S is a solution. Linear combinations of solutions are solutions. Odd, even, or periodic initial data produce solutions with the same symmetry in x.

Proposition 2.2. *(Maximum Absolute Value)*

$$|u(x,t)| \le M_\varphi + \frac{1}{2a} I_\psi,$$

where $M_\varphi = \|\varphi\|_\infty$ and $I_\psi = \max_{\alpha,\beta}\left|\int_\alpha^\beta \psi(y)\,\mathrm{d}y\right|$.

Definition 2.1. *(Interval of Dependence)*
The value $u(x_0,t_0)$ is determined by the initial data on the interval of dependence $[x_0 - at_0, x_0 + at_0]$, the base of the characteristic triangle with vertex (x_0,t_0).

Definition 2.2. *(Domain of Determination)*
Initial data on $[x_1,x_2]$ determine the solution throughout the domain of determination $\{x_1 + at \le x \le x_2 - at\}$.

Definition 2.3. *(Domain of Influence)*
A change of the data at x_0 affects the solution only in the domain of influence

$\{x_0 - at \le x \le x_0 + at\}$. A change of the data on $[x_1, x_2]$ affects the solution only in $\{x_1 - at \le x \le x_2 + at\}$.

Example.
On the half-line, consider

$$u_{tt} - a^2 u_{xx} = 0 \quad (x > 0, t > 0),$$

with $u(x,0) = \varphi(x)$, $u_t(x,0) = \psi(x)$, and $u(0,t) = 0$. Extend φ and ψ oddly to $\mathbb{R}$ and solve the Cauchy problem on the line. Restricting to $x \ge 0$ yields d'Alembert's formula for $x \ge at$, and

$$u(x,t) = \frac{-\varphi(at - x) + \varphi(x + at)}{2} + \frac{1}{2a}\int_{at-x}^{x+at} \psi(y)\,\mathrm{d}y$$

for $0 < x \le at$. The same extension argument with even data treats the Neumann condition $u_x(0,t) = 0$.

Remark.
Equivalently, write $u = F(x - at) + G(x + at)$ and determine F and G from the initial conditions when $x \ge at$, then from the boundary condition together with the values on $x = at$ when $0 < x \le at$.

Example.
Let $k > 1$ and $f(0) = h(0)$. For $u_{tt} - u_{xx} = 0$ in $0 < t < kx$, with $u(x,0) = f(x)$, $u_t(x,0) = g(x)$, and $u = h$ on $t = kx$, d'Alembert's formula holds for $x \ge t$. In the region $x \le t \le kx$ the backward characteristic $x - t = \text{const}$ meets $t = kx$ rather than $t = 0$. Set $\lambda = \frac{1+k}{1-k}$. Then

$$u(x,t) = h\!\left(\frac{x-t}{1-k}\right) + \frac{1}{2}[f(x+t) - f(\lambda(x-t))] + \frac{1}{2}\int_{\lambda(x-t)}^{x+t} g(y)\,\mathrm{d}y.$$

Theorem 2.2. *(Duhamel's Principle)*
Let $w(x,t;\tau)$ solve

$$w_{tt} - a^2 w_{xx} = 0 \quad (t > \tau), \quad w(x,\tau;\tau) = 0, \quad w_t(x,\tau;\tau) = f(x,\tau).$$

Then $u_2(x,t) = \int_0^t w(x,t;\tau)\,\mathrm{d}\tau$ solves

$$u_{tt} - a^2 u_{xx} = f(x,t), \quad u(x,0) = 0, \quad u_t(x,0) = 0.$$

Proof.
Differentiating under the integral, $u_2(x,t) = \int_0^t w \, \mathrm{d}\tau$ satisfies $\partial_t u_2 = \int_0^t w_t \, \mathrm{d}\tau$ because $w(x,t;t) = 0$, and

$$\partial_{tt} u_2 = w_t(x,t;t) + \int_0^t w_{tt} \, \mathrm{d}\tau = f(x,t) + a^2 \partial_{xx} u_2.$$

Both u_2 and $\partial_t u_2$ vanish at $t = 0$. ■

Remark.
The integrand $w(\cdot, t; \tau)\, \mathrm{d}\tau$ is the wave produced by an impulse $f(\cdot,\tau)\,\mathrm{d}\tau$ in the velocity at time τ.

Theorem 2.3.
If $\varphi \in C^2(\mathbb{R})$, $\psi \in C^1(\mathbb{R})$, and f is C^1, the Cauchy problem $u_{tt} - a^2 u_{xx} = f$ with data φ, ψ has the solution

$$u(x,t) = \frac{\varphi(x-at) + \varphi(x+at)}{2} + \frac{1}{2a}\int_{x-at}^{x+at} \psi(y)\,\mathrm{d}y + \frac{1}{2a}\int_0^t \int_{x-a(t-\tau)}^{x+a(t-\tau)} f(y,\tau)\,\mathrm{d}y\,\mathrm{d}\tau.$$

The double integral is the integral of f over the characteristic triangle G with vertex (x,t) and base $[x-at, x+at]$.

2.2 Higher-Order Wave Equations

Definition 2.4. *(Spherical Mean)*
For h on $\mathbb{R}^3$, the spherical mean on the sphere $|y-x| = r$ is

$$\overline{h}(x,r) = \frac{1}{4\pi r^2} \iint_{|y-x|=r} h(y)\,\mathrm{d}S_y = \frac{1}{4\pi}\iint_{|\alpha|=1} h(x + r\alpha)\,\mathrm{d}\omega.$$

Lemma 2.1. *(Darboux Equation)*
If $h \in C^2$, then

$$\left(\partial_{rr} + \frac{2}{r}\partial_r\right)\overline{h}(x,r) = \Delta_x \overline{h}(x,r),$$

with $\overline{h}(x,0) = h(x)$ and $\partial_r \overline{h}(x,0) = 0$. The even extension of $\overline{h}$ in r remains C^2.

Proof.
Differentiating under the integral and applying the divergence theorem,

$$\partial_r \overline{h} = \frac{1}{4\pi r^2} \iiint_{|y-x|\le r} \Delta h(y)\,\mathrm{d}y = \frac{1}{r^2}\int_0^r \rho^2 \Delta_x \overline{h}(x,\rho)\,\mathrm{d}\rho.$$

Multiplying by r^2 and differentiating in r yields the Darboux equation. The conditions at $r=0$ follow by direct inspection, using that $\iint_{|\alpha|=1} \alpha_i \,\mathrm{d}\omega = 0$. ∎

Theorem 2.4. *(Kirchhoff's Formula)*
If $\varphi \in C^3(\mathbb{R}^3)$ and $\psi \in C^2(\mathbb{R}^3)$, the Cauchy problem $u_{tt} - a^2\Delta u = 0$ in $\mathbb{R}^3$ has a unique C^2 solution

$$\begin{aligned} u(x,t) &= \partial_t\big(t\overline{\varphi}(x,at)\big) + t\overline{\psi}(x,at) \\ &= \partial_t\left(\frac{1}{4\pi a^2 t}\iint_{|y-x|=at} \varphi(y)\,\mathrm{d}S_y\right) + \frac{1}{4\pi a^2 t}\iint_{|y-x|=at} \psi(y)\,\mathrm{d}S_y. \end{aligned}$$

Proof.
With $v = r\overline{u}$ as above,

$$\overline{u}(x,r,t) = \frac{(r+at)\overline{\varphi}(x,r+at) + (r-at)\overline{\varphi}(x,r-at)}{2r} + \frac{1}{2ar}\int_{r-at}^{r+at} y\overline{\psi}(x,y)\,\mathrm{d}y,$$

using the even extension in r. The first summand tends to $\partial_t(t\overline{\varphi}(x,at))$ as $r \to 0$. In the second, the integrand $y\overline{\psi}$ is odd, so the integral over the symmetric interval about the origin vanishes and the remainder tends to $t\overline{\psi}(x,at)$. Uniqueness follows because every C^2 solution arises in this way. ∎

Example.
For $\varphi(x) = x_1^2 + x_2^2 + x_3^2$ and $\psi(x) = x_1 x_2$, the radial problem for the first datum and the spherical mean of the second give

$$u(x,t) = x_1^2 + x_2^2 + x_3^2 + 3a^2t^2 + x_1x_2t.$$

Remark.
Writing $S(t)\psi := t\overline{\psi}(x,at)$ for the solution with vanishing displacement, Kirchhoff's formula is $u = \partial_t(S(t)\varphi) + S(t)\psi$. Differentiating the spherical integral produces the equivalent form

$$u(x,t) = \frac{1}{4\pi a^2 t^2}\iint_{|y-x|=at} \left(\varphi(y) + \nabla\varphi(y)\cdot(y-x) + t\psi(y)\right)\mathrm{d}S_y.$$

The value $u(x,t)$ depends on the data on the sphere $|y-x|=at$. The domain of influence of a point $(x,0)$ is the characteristic cone $|y-x| = a\tau$ for $\tau > 0$.

Remark.
In $\mathbb{R}^n$ the spherical mean is $\overline{h}(x,r) = \frac{1}{\omega_n}\iint_{|\alpha|=1} h(x+r\alpha)\,\mathrm{d}\omega$, with surface area $\omega_n = n\frac{\pi^{\frac{n}{2}}}{\Gamma(\frac{n}{2}+1)}$. It satisfies the Darboux equation

$$\left(\partial_{rr} + \frac{n-1}{r}\partial_r\right)\overline{h} = \Delta_x\overline{h},$$

and $\overline{u}$ solves the Euler–Poisson–Darboux equation

$$\overline{u}_{tt} - a^2\left(\overline{u}_{rr} + \frac{n-1}{r}\overline{u}_r\right) = 0.$$

Equivalently, $(r\overline{u})_{tt} - a^2(r\overline{u})_{rr} = (n-3)\partial_r\overline{u}$. For $n=3$ the right-hand side vanishes. For odd $n>3$ repeated application of $r^2\partial_r$ reduces the equation to the one-dimensional wave equation. For $n=5$ the combination $w = r^2\partial_r\overline{u} + 3r\overline{u}$ satisfies $w_{tt} - a^2 w_{rr} = 0$, and $u(x,t) = \lim_{r\to 0}\frac{w}{3r}$. For $n=2$ the reduction is not available, and one uses descent instead.

Theorem 2.5. *(Poisson's Formula)*
If $\varphi \in C^3(\mathbb{R}^2)$ and $\psi \in C^2(\mathbb{R}^2)$, the Cauchy problem in $\mathbb{R}^2$ has a unique C^2 solution

$$u(x,t) = \frac{1}{2\pi a}\partial_t \iint_{|y-x|\le at} \frac{\varphi(y)}{\sqrt{a^2t^2 - |y-x|^2}}\,\mathrm{d}y + \frac{1}{2\pi a}\iint_{|y-x|\le at} \frac{\psi(y)}{\sqrt{a^2t^2-|y-x|^2}}\,\mathrm{d}y.$$

Proof.
On $|y-x| = at$ one has $y_3 - x_3 = \pm\sqrt{a^2t^2 - |y'-x|^2}$. The two hemispheres contribute

$$\iint_{|y-x|=at} \varphi(y_1,y_2)\,\mathrm{d}S_y = 2at \iint_{|y-x|\le at} \frac{\varphi(y)}{\sqrt{a^2t^2-|y-x|^2}}\,\mathrm{d}y.$$

Substituting into Kirchhoff's formula yields the stated coefficients. The integrals do not depend on x_3. ■

Theorem 2.6. *(Decay)*
Let $n = 3$, and suppose $\varphi \in C_0^1(\mathbb{R}^3)$ and $\psi \in C_0(\mathbb{R}^3)$ vanish outside $\overline{B}(0,\rho)$ and are bounded together with $\nabla\varphi$. Then $|u(x,t)| \le Ct^{-1}$ for $t \ge 1$, uniformly in x. In general,

$$|u(x,t)| \le C(1+t)^{-\frac{n-1}{2}}.$$

Proof.
On the sphere $|y-x| = at$ the integrand of the rewritten Kirchhoff formula is supported in a set of area at most $4\pi\rho^2$. The factors φ and $\nabla\varphi\cdot(y-x)$ contribute $O(1)$ and $O(t)$, which together with the prefactor $\frac{1}{t^2}$ yield $O(t^{-1})$. ■

Theorem 2.7. *(Retarded Potentials)*
In three dimensions,

$$u_2(x,t) = \frac{1}{4\pi a^2}\iiint_{|y-x|\le at} \frac{f\!\left(y, t-\frac{|y-x|}{a}\right)}{|y-x|}\,\mathrm{d}y.$$

In two dimensions,

$$u_2(x,t) = \frac{1}{2\pi a}\int_0^t \iint_{|y-x|\le a(t-\tau)} \frac{f(y,\tau)}{\sqrt{a^2(t-\tau)^2-|y-x|^2}}\,\mathrm{d}y\,\mathrm{d}\tau.$$

Proof.
For $n = 3$, Kirchhoff's formula for w is

$$w(x,t;\tau) = \frac{1}{4\pi a^2(t-\tau)}\iint_{|y-x|=a(t-\tau)} f(y,\tau)\,\mathrm{d}S_y.$$

Integrating in τ and substituting $r = a(t-\tau)$ converts the integral over spherical shells into the retarded volume potential. The two-dimensional identity is Poisson's formula integrated in τ. ■